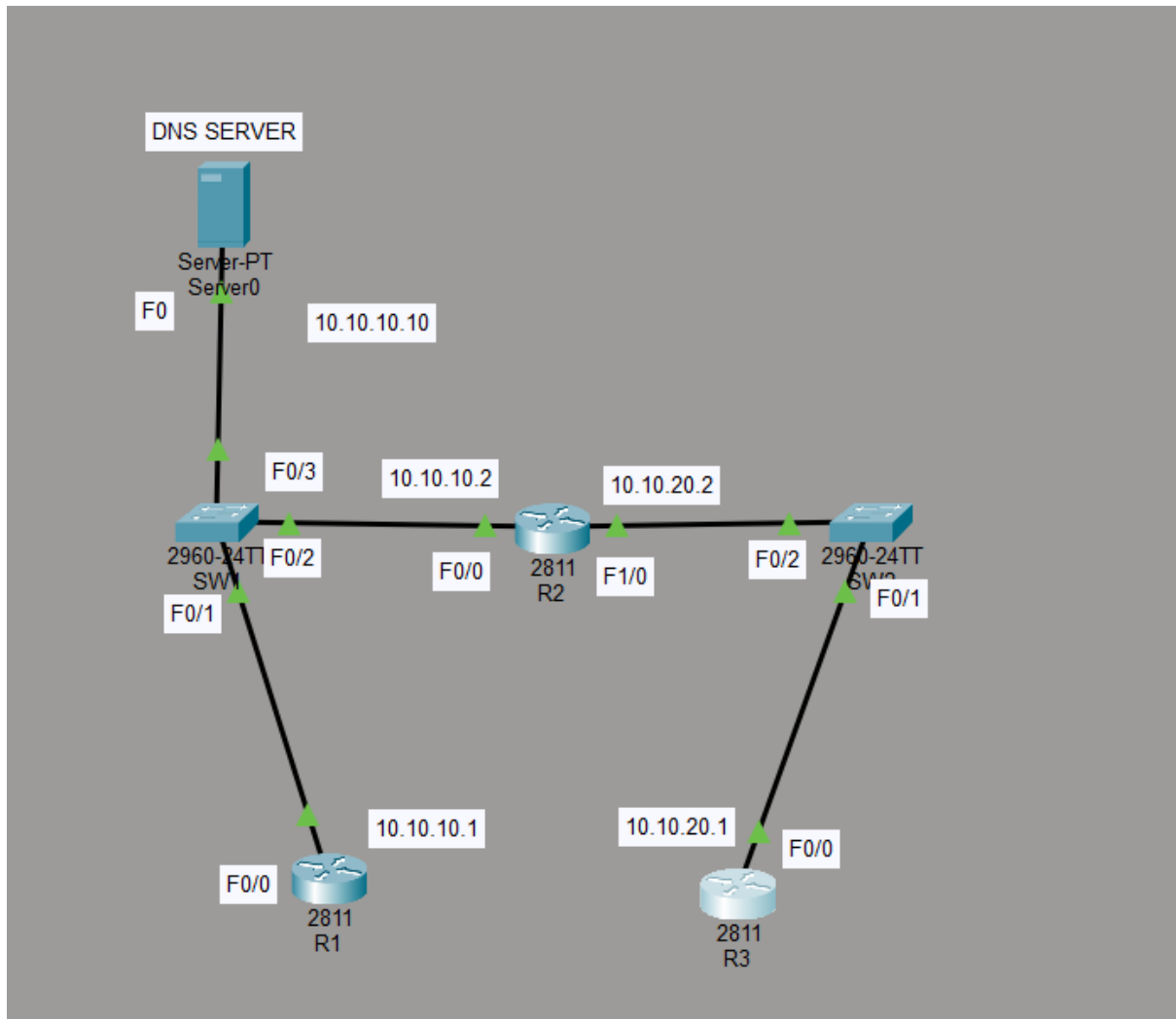


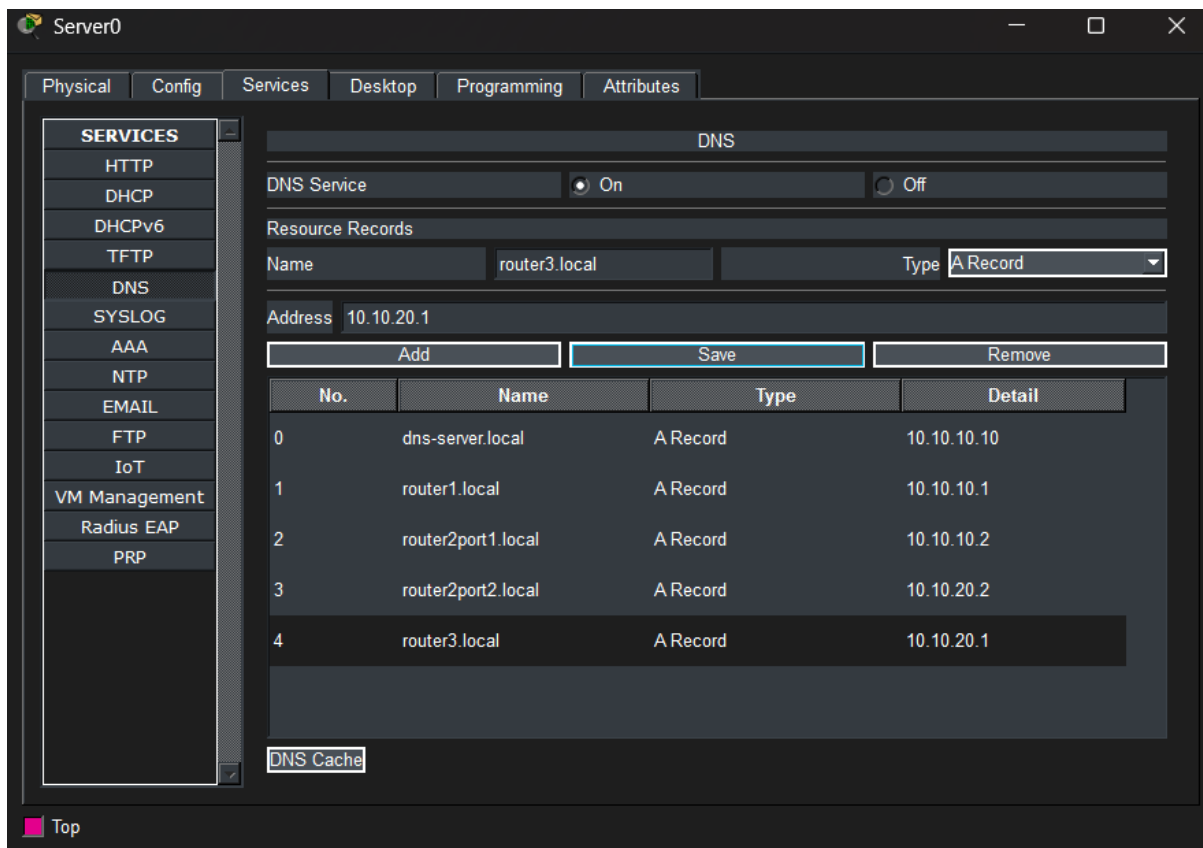
Topology



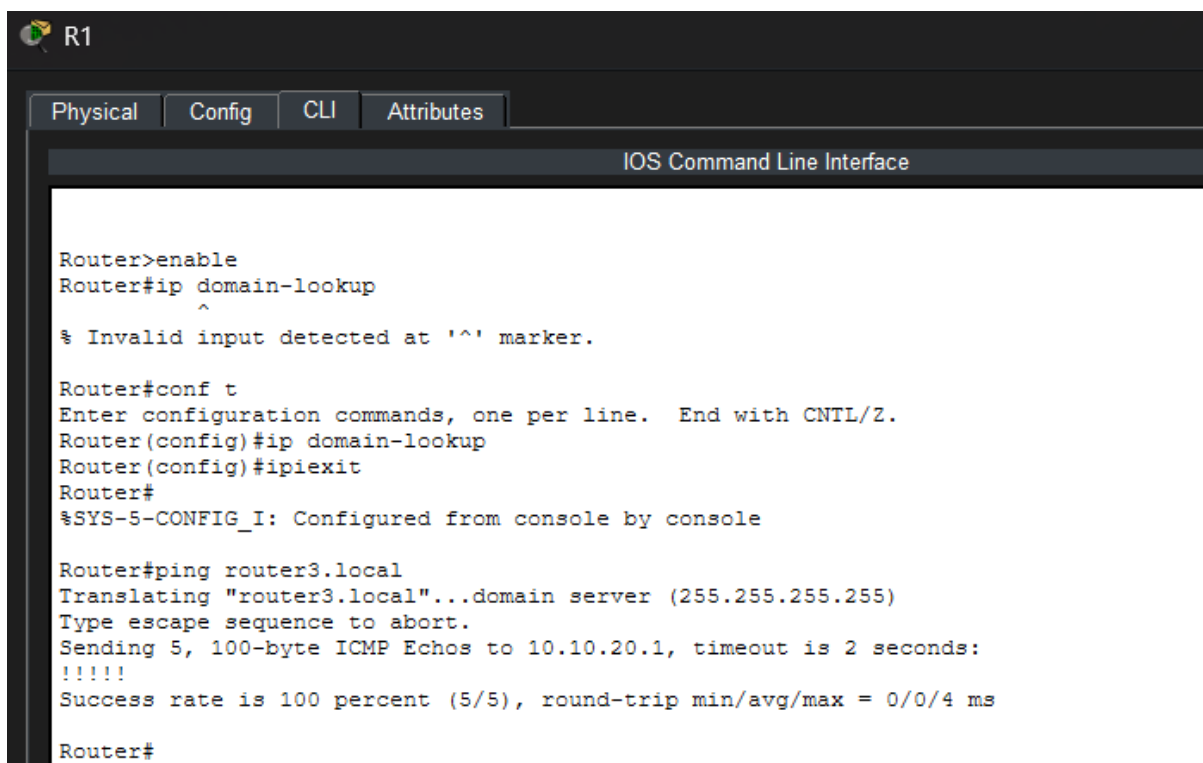
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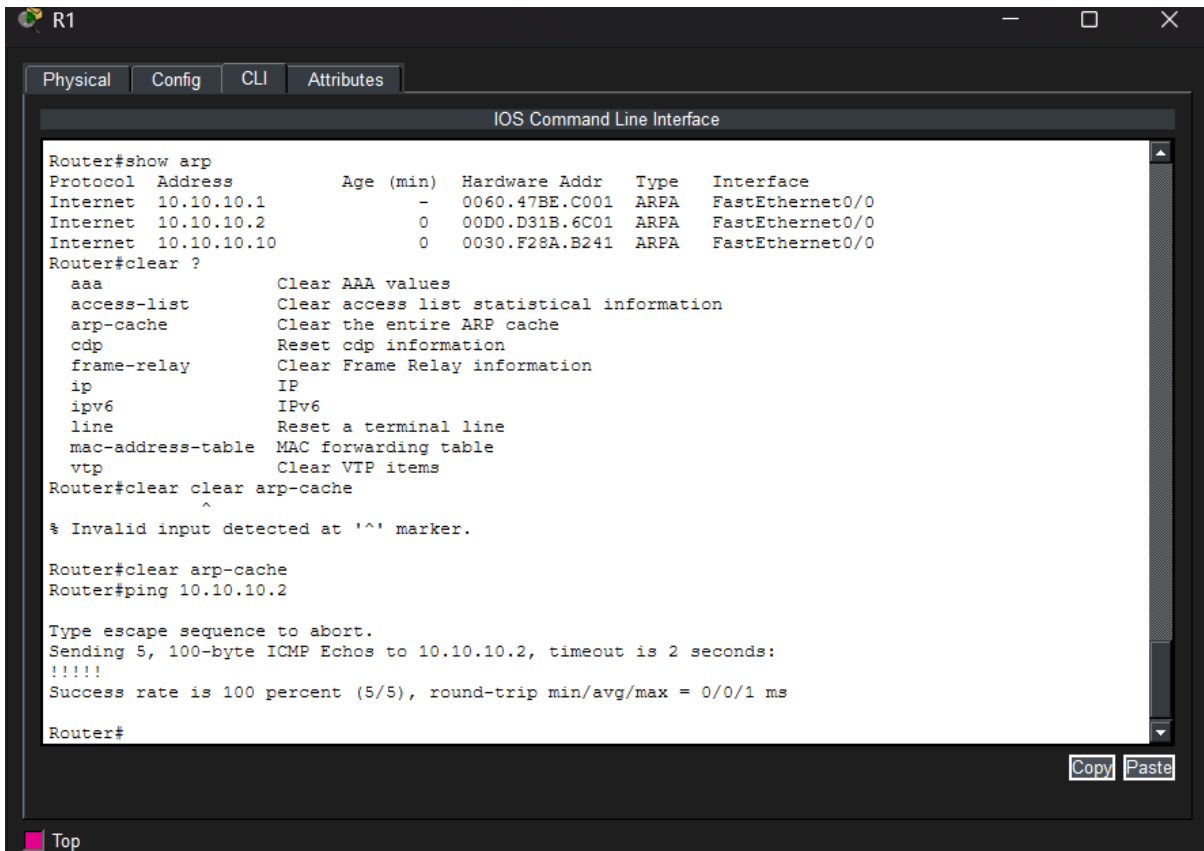
Step 16: Configuring routers as DNS clients



Step 17: Ping a hostname instead of an ip address



Step 18: Clearing the ARP cache on Router 1, then pinging a directly connected neighbour
(Router 2 10.10.10.2)



```
Router#show arp
Protocol Address      Age (min)  Hardware Addr  Type   Interface
Internet 10.10.10.1          -    0060.47BE.C001  ARPA   FastEthernet0/0
Internet 10.10.10.2          0    00D0.D31B.6C01  ARPA   FastEthernet0/0
Internet 10.10.10.10         0    0030.F28A.B241  ARPA   FastEthernet0/0
Router#clear ?
    aaa                Clear AAA values
    access-list         Clear access list statistical information
    arp-cache           Clear the entire ARP cache
    cdp                 Reset cdp information
    frame-relay         Clear Frame Relay information
    ip                  IP
    ipv6                IPv6
    line                Reset a terminal line
    mac-address-table   MAC forwarding table
    vtp                 Clear VTP items
Router#clear clear arp-cache
      ^
% Invalid input detected at '^' marker.

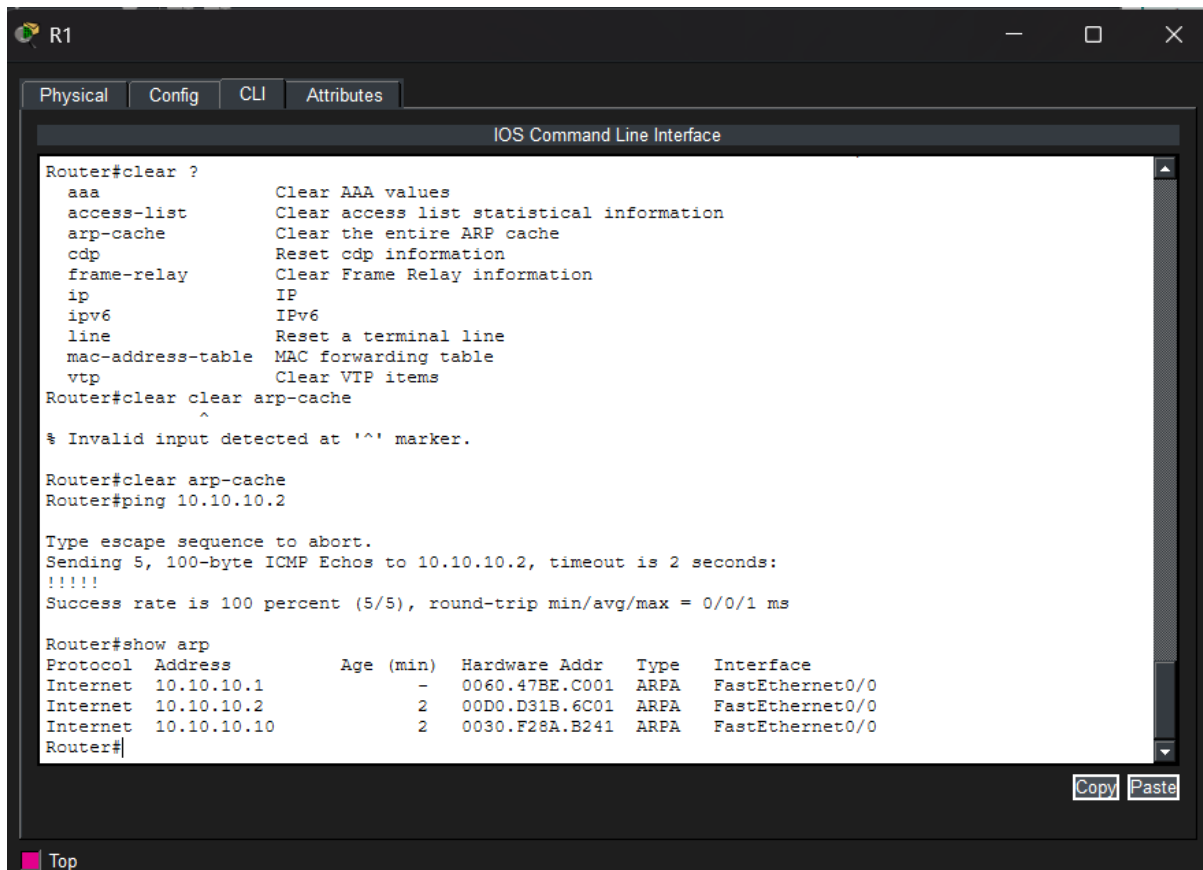
Router#clear arp-cache
Router#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router#
```

Step 19: Examining ARP caches on router

Router 1



```
R1
Physical Config CLI Attributes
IOS Command Line Interface

Router#clear ?
  aaa                Clear AAA values
  access-list        Clear access list statistical information
  arp-cache          Clear the entire ARP cache
  cdp                Reset cdp information
  frame-relay        Clear Frame Relay information
  ip                 IP
  ipv6               IPv6
  line               Reset a terminal line
  mac-address-table  MAC forwarding table
  vtp                Clear VTP items
Router#clear clear arp-cache
      ^
% Invalid input detected at '^' marker.

Router#clear arp-cache
Router#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router#show arp
Protocol Address          Age (min)  Hardware Addr  Type   Interface
-----
Internet 10.10.10.1      -         0060.47BE.C001  ARPA   FastEthernet0/0
Internet 10.10.10.2      2         00D0.D31B.6C01  ARPA   FastEthernet0/0
Internet 10.10.10.10     2         0030.F28A.B241  ARPA   FastEthernet0/0
Router#
```

The age of the address it cached is now 2 minutes in age, aka when I did the ping, so it identified my ping to router 2.

Reflection questions

20: An ARP entry gives you information on the local subnet and its MAC address. While a routing table shows you the type and extra info on the connection its mapped.

21: The ARP lookup finds the destination MAC address and ensure it is still the correct recipient. It also shows the next hop, not just where the final destination is, it helps the route itself.